EXP NO: 4
17/2/26

Design a regulated power supply circuit using a voltage regulator IC.

AIM:

To design a regulated power supply circuit using a voltage regulator IC, simulate the circuit in LT spice, and test its performance for different load conditions.

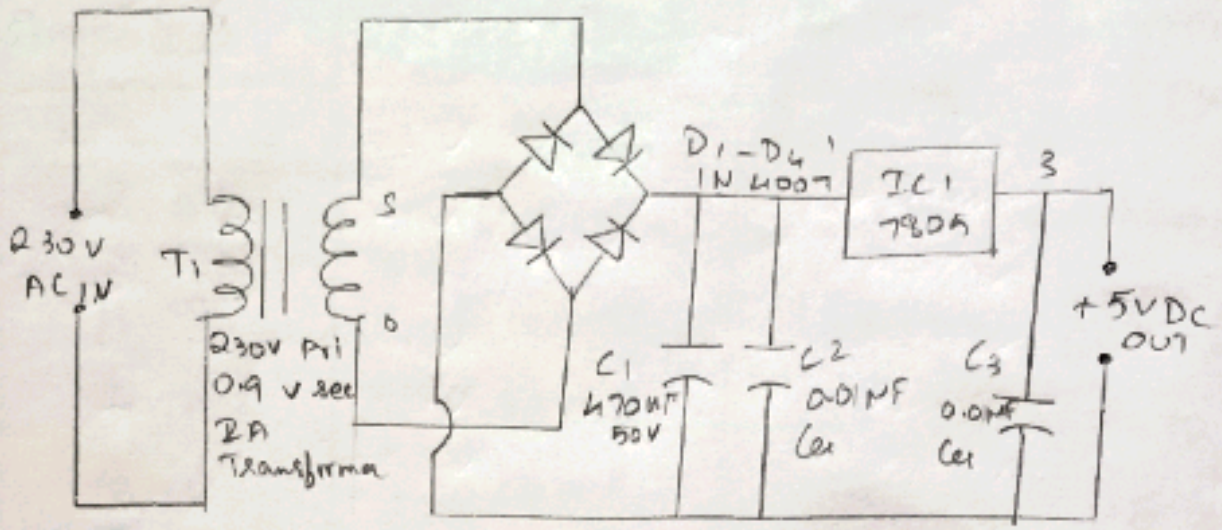
Software required:

-> LTSpice simulator.

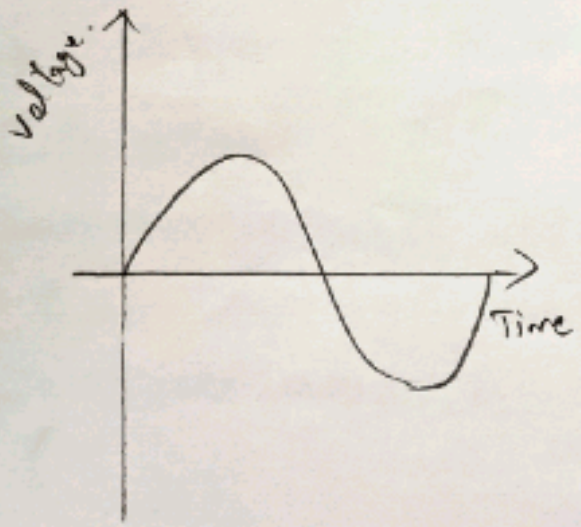
PROCEDURE.

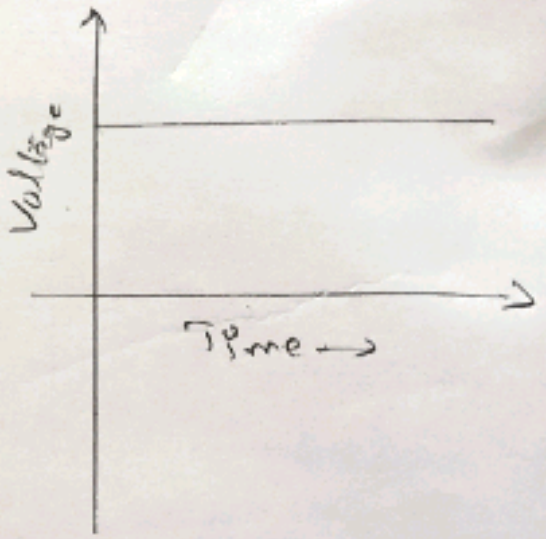
1) Connecting the circuit on bread board as per the circuit diagram.
2) Connect the primary of the transformer to main supply i.e. 230V, 50Hz.
3) Connect the primary elements and complete the circuit.
4) Measure the output by varying the input load of various values.
5) Compare the output for different load values.

Circuit Diagram.



Model Graph.





TABULATION:

Sl NO	Input Voltage (V_{in})	Input frequency (Hz)	Output voltage (V_o)
1	6V	50Hz	5.0V
2	7V	50Hz	5.0V
3	8V	50Hz	5.0V
4	9V	50Hz	5.0V
5	10V	50Hz	5.0V
6	11V	50Hz	5.0V
7	12V	50Hz	5.0V
8	13V	50Hz	5.0V
9	14V	50Hz	5.0V
10	15V	50Hz	5.0V

RESULT:

Thus a regulated power supply circuit using a voltage regulator IC was designed and the circuit was simulated using LT Spice and its performance for different load condition was tested.